

Frequency-Coverage Limits in Assessing Gravity Signals from Oceanic Mass Redistribution

[PENDING: author list and affiliations]

Reproducible results draft; author metadata pending

Abstract

Oceanic mass redistribution produces direct gravitational attraction and, for surface loads, elastic/potential and height responses over timescales from seconds to months. We develop an evidence-bounded process–distance–instrument framework with explicit source geometry, observation duration, spectral coverage, and instrument-specific noise. The primary six-process production ensemble contains 1,446 records of direct radial gravity; elastic and height components are retained separately where evaluated and are not implied for every process branch. None passes a predeclared 90% signal-energy coverage gate for the three admitted vertical-gravity literature anchors: tide, storm surge, eddy and internal-wave signals have no overlapping band, whereas tsunami and landslide records have only partial coverage. We therefore withhold SNR classifications rather than extrapolating instrument curves. An independent open-data Helgoland calculation reproduces the published BSH gravity-to-height ratio within 15.6%. The framework identifies frequency coverage, source geometry and mass compensation as separate limits on defensible ocean-gravity detectability statements.

1 Introduction

Environmental Newtonian-noise studies already quantify atmospheric, seismic, and ocean-related gravitational disturbances for gravitational-wave facilities; they are not instrument-noise calculations developed here. Likewise, terrestrial gravimetry has detected non-tidal ocean loading during storm surges. The present increment is a cross-process, cross-distance, cross-instrument assessment under one auditable convention. It first asks whether available instrument evidence covers enough signal energy to authorize a detectability decision. It does not claim that the underlying gravity kernels, loading theory, or individual instrument concepts are new.

2 Methods

2.1 Observables and sign conventions

All kernels use SI units. For a surface mass anomaly $\Delta\sigma$,

$$\Delta\mathbf{g}_{\text{direct}}(\mathbf{r}, t) = G \int_A \Delta\sigma(\mathbf{x}, t) \frac{\mathbf{x} - \mathbf{r}}{|\mathbf{x} - \mathbf{r}|^3} dA. \quad (1)$$

For a volume-density anomaly ρ' , the analogous integral is evaluated over volume. The gravity-gradient convention is $T_{ij} = \partial g_i / \partial x_{\text{observation}, j}$. Direct attraction, combined elastic gravity and vertical displacement are stored separately. The LoadDef CE gravity term combines deformation and potential response according to its source convention and is never represented as direct attraction.

2.2 Earth, grid and coastal geometry

Reference calculations include local planar, mean-radius spherical, and WGS 84 ellipsoidal geometry. Spherical and ellipsoidal cell areas are integrated in their native surface coordinates. Missing ocean values, hard masks, and partial coastal-cell fractions are explicit. Grid refinement and domain truncation are tested before a dataset is admitted to a result.

For idealized compact surface anomalies evaluated against terrestrial instruments, source cells and the station lie on a spherical Earth. Distance is the great-circle surface standoff from the represented three-standard-deviation source support. Compact anomalies translate parallel to an idealized coastline, preventing a source cell from coinciding with the station. A rejected local-plane near-field calculation is retained in the experiment preregistration history: its midpoint quadrature exceeded the continuous-sheet amplitude bound and did not enter the registered output.

2.3 Ocean-process models

The six process families are summarized in Table 1. Surface and compensated three-dimensional variants are retained only where physically supported. Engineering fixtures used for software validation are excluded from scientific uncertainty statements. Parameter values are linked to named events, catalogue means or published ranges. They define sensitivity designs, not occurrence probabilities or population percentiles. The mesoscale-eddy Gaussian branch uses catalogue-scale amplitude, radius and propagation speed from Chelton et al. [2]. Surface-wave packets and density dipoles are mass balanced before gravity integration.

Table 1: Primary direct-gravity representations in P1-E006. Record counts include repeated distances, durations or paired sensitivity points and are not counts of independent natural events.

Process	Evidence/representation		Independent physical family	Records
Tide	1-m M2 spherical-disk transfer		one geometry diagnostic; 5 distances $\times$ 32 durations	160
Storm surge	gridded 2022 Helgoland field	BSH	one named model case	1
Eddy	catalogue-mean Gaussian SSH	translating	one representative mean case; 5 distances	5
Internal wave	compensated density dipole		one model family; 128 paired sensitivities $\times$ 5 distances	640
Tsunami	mass-balanced packet	Gaussian	one packet family; 96 paired sensitivities $\times$ 5 distances	480
Landslide	Storegga conserved relocation		one named extreme family; 32 paired sensitivities $\times$ 5 distances	160

2.4 Spectra and detectability

For transient signals, expected matched-filter signal-to-noise ratio is

$$\rho^2 = 4 \int_{f_{\min}}^{f_{\max}} \frac{|\tilde{s}(f)|^2}{S_n(f)} df. \quad (2)$$

Fourier normalization, detrending, one-sided PSD, and observation duration are versioned. A classification is forbidden unless the instrument curve covers at least 90% of signal spectral energy. Vertical gravity, gravity gradient, and gravitational-wave equivalent strain remain different observables and are never merged into one sensitivity curve. The admitted vertical-gravity entries are an iGrav quiet self-noise anchor and short-term AQG-A01 and FG5#228 anchors; their declared frequency intervals are not extrapolated.

2.5 Parameter design and uncertainty

Latin-hypercube and stratified designs are labelled as scenario envelopes unless every dimension has a source-supported probability distribution. Engineering ranges cannot be reported as probabilistic 5–95% intervals. Density-compensated models retain cancellation and far-field suppression rather than replacing them with equivalent uncompensated water columns. Source standoffs are 10, 30, 100, 300 and 1,000 km, and every coverage-limited outcome is retained.

3 Validation

Analytic tests cover point masses, finite disks, rectangles, Gaussian sheets, volume cells, gradient symmetry and trace, WGS 84 area, mass conservation, grid convergence, domain truncation, and time/frequency normalization. The complete software audit contains 461 passing tests and 12 passing subtests after the structural-closure implementation.

The LoadDef v1.2.2 PREM/CE provider reproduces all 50 published benchmark rows used by the project for angular distance, radial displacement and combined elastic gravity [3]. Direct attraction is calculated independently; LoadDef’s Newtonian term is not added a second time.

The Helgoland model-only reproduction uses 242 BSH-HBMnoku files and 720 hourly fields conservatively remapped to a 1/8-degree WGS 84 grid. Direct attraction, combined elastic gravity and vertical displacement remain separate. After the registered sign conversion at the literature-comparison boundary, the computed gravity-to-height ratio is $-2.266 \text{ nm s}^{-2} \text{ mm}^{-1}$, compared with $-2.684 \text{ nm s}^{-2} \text{ mm}^{-1}$ reported by Voigt et al. [5]. The 15.6% difference passes the predeclared 20% tolerance for replacing SPOTL/Gutenberg–Bullen and a three-year tidal analysis with LoadDef PREM/CE and the registered 30-day harmonic fit.

4 Results

Figure 1 compares process timescales with the admitted instrument-curve support. M2 tide and the M2 internal-wave branch lie near $2.24 \times 10^{-5} \text{ Hz}$. Storm-surge and eddy ranges are lower. The three admitted vertical-gravity anchors begin at 10^{-3} Hz . Tsunami characteristic content crosses that edge, while the Storegga transition lies mainly below it; neither has 90% of signal energy inside a complete admitted band.

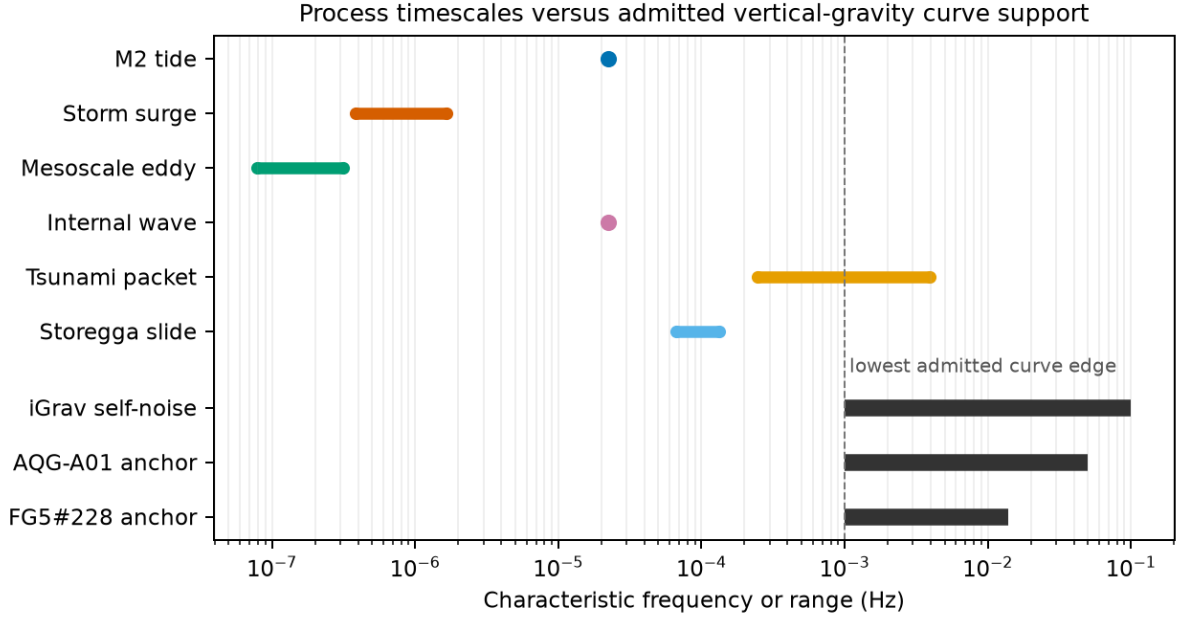


Figure 1: Characteristic process frequencies and admitted vertical-gravity curve support. Points denote fixed constituent frequencies and horizontal bars denote sensitivity ranges. Curve limits are not extrapolated.

The resulting coverage matrix is shown in Figure 2. All 806 long-period records have no covered spectral interval for any admitted curve. All 640 tsunami and Storegga records have some overlapping bins but fail the 90% energy criterion. Consequently no production record is labelled detectable or undetectable. This differs from a statement of zero SNR: available curve support is insufficient to make the comparison.

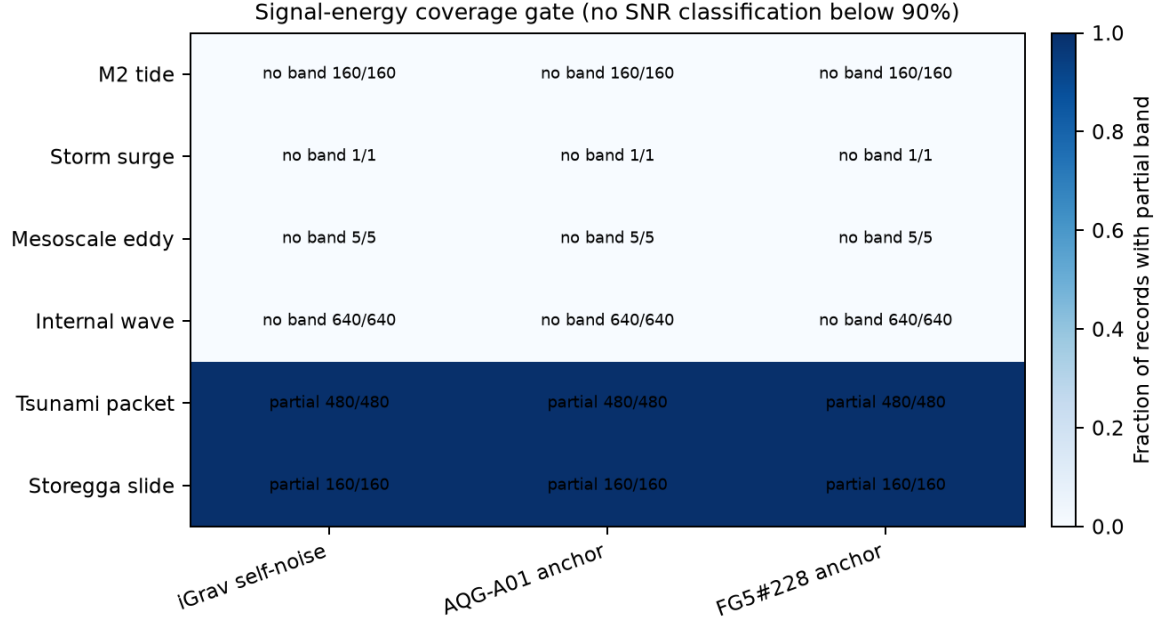


Figure 2: Process–instrument signal-energy coverage. “No band” means fewer than two positive-frequency bins overlap a curve; “partial” means overlap is present but covers less than 90% of signal energy.

Direct radial-gravity amplitudes decay with spherical surface standoff (Figure 3). The Storegga named-event branch has the largest primary-branch envelope, with median peak amplitude decreasing from about $9.5 \times 10^{-9} \text{ m s}^{-2}$ at 10 km to $1.0 \times 10^{-10} \text{ m s}^{-2}$ at 1,000 km. The compensated tsunami envelope decreases from approximately $3.1\text{--}6.5 \times 10^{-10}$ to $4.9\text{--}12.7 \times 10^{-12} \text{ m s}^{-2}$ across the same standoffs. The internal-wave dipole has an off-axis maximum near 100 km, illustrating why amplitude need not decrease monotonically inside a compensated source’s near field.

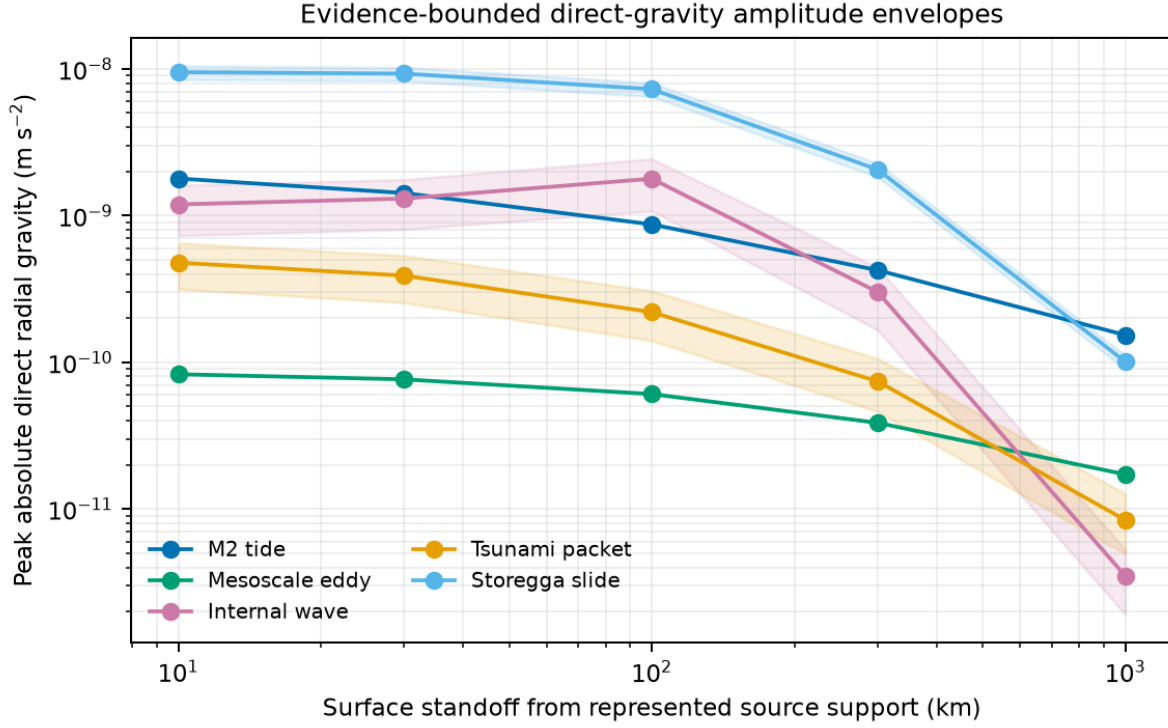


Figure 3: Peak absolute direct radial gravity versus surface standoff. Lines show the median sensitivity design and shading spans the sampled minimum and maximum; shading is not a probability interval.

Figure 4 separates the density/displacement, wave-packet and slide-parameter envelopes. Every tsunami crest and trough pair cancels signed surface mass to numerical tolerance, and every slide record relocates rather than creates solid mass.

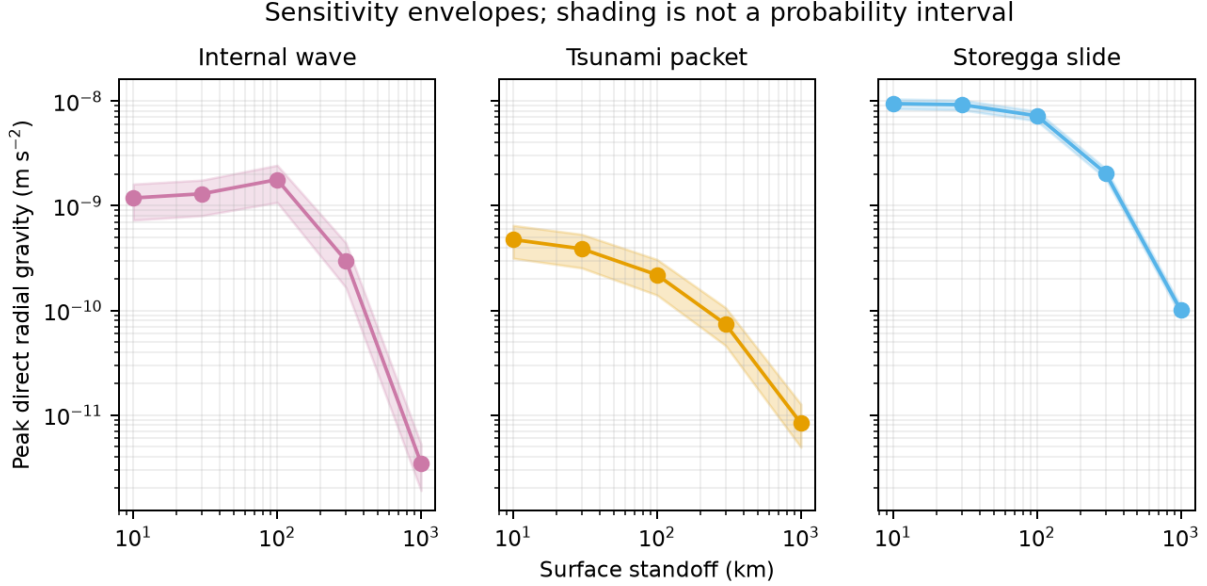


Figure 4: Evidence-bounded parameter sensitivity for three primary branches. The bands represent design extrema and must not be read as confidence or credible intervals.

A separate registered structural-closure experiment maps the $3.0 \times 10^{15} \text{ J}$ Tohoku propagated-energy estimate (6% uncertainty) reported by Tang et al. [4] to the same idealized mass-balanced Gaussian packet. Under the declared linear-long-wave convention $E = \rho g \int \eta^2 dA$, the 300–400 km separation sensitivity gives a central crest amplitude of 4.359 m and an energy-uncertainty range of 4.226–4.488 m. This branch is deliberately separate from the 1.0–1.8 m deep-ocean amplitude-normalized branch; their difference measures normalization/model-form sensitivity and is not combined as a probability interval.

5 Discussion

The principal result is a boundary between classified, coverage-limited, and physically suppressed signals. Compensated internal structures can decay much faster than surface-load estimates, while long-period processes may be strong but unclassifiable when empirical noise curves do not cover their frequency band. An amplitude–noise intersection is not a sufficient detectability test: observation duration and spectral energy coverage must be evaluated first.

The absence of a production SNR classification is therefore informative. It identifies a data requirement—site and instrument noise curves extending into the process band—rather than an invitation to extend short-term white-noise anchors by assumption. Environmental Newtonian-noise calculations expressed as gravitational-wave strain are likewise not interchangeable with gravimeter self-noise [1].

The Helgoland reproduction supplies an independent positive validation of the elastic loading scale but not of observation/model correlation. The reported 85 nm s^{-2} and -34 mm extrema are observations, not BSH model acceptance targets. Reproducing the reported correlation and RMS reduction would require authenticated iGrav047 data and a paper-equivalent processed 2022 GNSS series.

6 Conclusions

The current public instrument anchors do not support a decision-grade cross-process detectability map: no production record reaches the required signal-energy coverage. This result does not imply absent physical signals or universal non-detectability. It establishes the frequency support that future site-specific noise measurements must provide, while the direct-gravity scenario suite quantifies how geometry, distance and mass compensation alter the source signal. Total terrestrial observables require separately validated elastic/potential and height components and are not inferred from the direct atlas alone.

7 Data and code availability

Code, compact experiment metadata, checksums, figure inputs and manuscript sources are maintained in the public project repository. Bulk BSH NetCDF and GNSS RINEX inputs remain on the compute data volume and are represented by source queries, inventories and checksums. Restricted gravimeter data and large gridded products remain outside Git and are represented by versioned manifests, access conditions, time ranges, and checksums.

8 Limitations

The production atlas contains one primary branch per process family, plus the energy-normalized tsunami sensitivity. Remaining structural branches are explicit scientific exclusions, not hidden omissions or zero-effect assumptions. The eddy design cannot simultaneously hold peak SSH, radius, and positive mass fixed across Gaussian and compact quadratic profiles; the compensated eddy lacks an SSH-to-density-profile mapping. Continuous mode-1 and free-surface internal-wave branches lack a boundary-conditioned vertical profile and coupling coefficient. Mediterranean/Storfjorden slide evidence does not jointly close amplitude and duration, and the generated-water-wave branch lacks an event-specific generation/entrainment transfer model.

The admitted instrument curves are literature anchors rather than complete site-specific environmental noise models. The iGrav curve estimates quiet self-noise, AQG and FG5 entries are short-term anchors, and none authorizes low-frequency extrapolation. Thus the matrix establishes a coverage boundary, not universal instrument performance. Haikou observation reproduction remains author-restricted; this does not affect the six-process ensemble. A PEGS simulator is unrelated to Paper 1. Final authorship, affiliations, target-journal format and the decision to include unavailable observation rows require human author approval.

References

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